

Motivation

Accurate current measurement is fundamental to the development and validation of power electronic converters and electric drive systems. The measured current is often directly used for protection, closed-loop control, estimation approaches, and experimental validation. Therefore, even a small measurement error can propagate into the control system and lead to incorrect conclusions about the behaviour of the converter or drive.

In practical implementations, however, a current sensor does not always produce the ideal output predicted by its datasheet. The actual measurement chain may contain errors arising from sensor tolerances, signal-conditioning circuits, level shifters, ADC scaling, offsets, component tolerances, temperature, and PCB implementation.

This project investigates the calibration of an ACS772 current sensor together with its associated signal processing stage. Instead of relying blindly on the nominal sensor characteristics, experimentally measured current values were compared against a reference current measurement. A linear calibration model was then obtained from the experimental data.

Objective

The primary objective of this work was to determine an experimentally validated relationship between the output of the current-sensing circuit and the actual current.

The calibration was performed by assuming a linear relationship of the form

$$I_{\text{actual}} = mI_{\text{sensor}} + c$$

where I_{sensor} is the current obtained from the sensor measurement chain, I_{actual} is the reference current, m is the calibration gain, and c is the calibration offset.

The experimentally obtained values of m and c were subsequently used to reconstruct the calibrated current waveform.

Calibration Methodology

The experimental data consisted of two current measurements recorded simultaneously:

- **Reference current:** current measured using a calibrated or reference current measurement system.
- **Sensor current:** current obtained from the ACS772-based sensing and level-shifting circuit.

The recorded data were imported into MATLAB and divided into consecutive data windows. For each window, the sensor measurement was considered as the independent variable and the reference measurement as the dependent variable.

For each data window, a first-order polynomial fit was performed:

$$I_{\text{actual}} = m_i I_{\text{sensor}} + c_i$$

The slope m_i and intercept c_i obtained from each window were stored. The final calibration coefficients were obtained by averaging the individual estimates:

$$m = \frac{1}{N} \sum_{i=1}^N m_i$$

$$c = \frac{1}{N} \sum_{i=1}^N c_i$$

The resulting calibration equation was then applied to the complete sensor waveform:

$$I_{\text{calibrated}} = m I_{\text{sensor}} + c$$

This approach makes the calibration dependent on the behaviour of the actual measurement hardware rather than solely on nominal component parameters.

MATLAB Implementation

A MATLAB script was developed to automate the calibration process. The implementation performs the following operations:

1. Imports the oscilloscope/current-sensor data.
2. Separates the reference and sensor current measurements.
3. Divides the recorded data into successive calibration windows.
4. Performs linear regression for each window.
5. Extracts the slope and intercept for each fit.
6. Calculates the mean calibration coefficients.
7. Applies the resulting calibration equation to the complete sensor waveform.
8. Compares the calibrated waveform with the reference waveform.

For each calibration window, the linear regression is expressed as

$$I_{\text{actual},i} = m_i I_{\text{sensor},i} + c_i$$

The final calibrated current is therefore obtained directly from the experimentally determined mean coefficients.

Conclusion

An experimental calibration procedure was developed for an ACS772-based current measurement system with an external gain of 0.5. The sensor was used to measure current through a resistive load tied to an ac voltage source. The resulting calibration equation,

$$I_{\text{actual}} = 19.3854 I_{\text{sensor}} - 24.0561$$

was applied to the measured waveform and successfully reproduced the reference current waveform as shown in Figure 1.

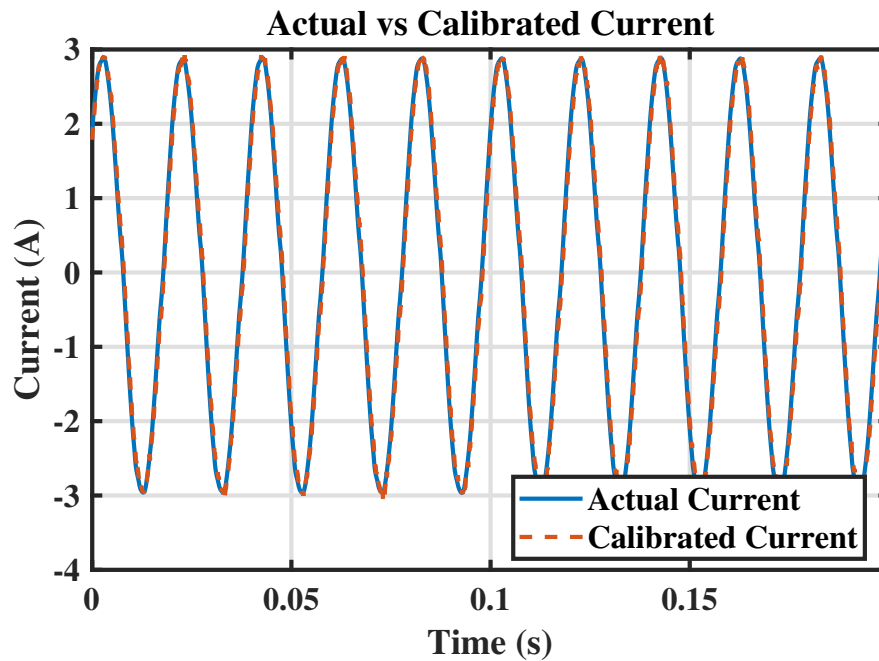


Figure 1: Comparison between the reference current and the experimentally calibrated sensor current.

Comparison with the Nominal Sensor Model

For the ACS772-based measurement system used in this experiment, the nominal sensor output was considered to vary from 0 V at -20 A to 5 V at $+20$ A. The corresponding sensor equation is

$$V_s = 0.125I + 2.5$$

An additional signal-conditioning stage with a gain of 0.5 was then included. Hence, the final sensor-chain output is

$$V_o = 0.5V_s = 0.0625I + 1.25$$

Expressing the current as a function of the measured output gives the nominal relationship

$$I_{\text{nom}} = 16V_o - 20$$

However, experimental calibration using the reference current sensor resulted in

$$I_{\text{cal}} = 19.3854V_o - 24.0561$$

Thus, the experimentally obtained slope is

$$\frac{19.3854 - 16}{16} \times 100 \approx 21.16\%$$

The experimentally obtained transfer characteristic therefore differs noticeably from the nominal model. This highlights the importance of calibrating the complete sensing chain, including the current sensor, signal-conditioning circuit, and data-acquisition system, rather than blindly relying on the nominal sensor relationship.